

TASK 3 : AI Use case Evaluation

1. Auto code documentation and review

Problem: Normally human reviewer may sometimes miss some lines , bugs or security issues, or style of writing code(ex : microsoft may have different code writing style than google). We need ai model who does this job.

Data required: Historical or past PRs or past codebase (ex: A company may use its past codebases and make the model use to the codebase so that it functions according to the organization style) , review comments, security or incident reports (to learn and understand what bad code actually caused problem) Readme or guides.

AI type: Genai (LLM for reasoning about code) , traditional static analysis and rule-based or ai workflows . Sometimes pure and raw LLM output is not enough to trust alone for things like security.

Validation: Precision recall against a labeled set of known historical bugs/vulnerabilities, AB testing.

2. Knowledge Retrieval from Engineering Documents [my Major project]

Problem: people working in Oil and gas industries often spend a lot of time searching through PDFs, manuals, government laws and legislative policies and previous reports to find resources and areas for their oil and gas operations . Instead of searching manually, they can ask questions in natural language and receive answers with references to the documents like News articles , laws published by the government , policies etc.

Data required:Government policies for land laws and legislatives , documentation, previous project reports, design specifications, and internal organisation knowledge base.

AI type: Genai (NLP) with RAG.

Validation: Test the system using a set of questions (like can i explore this areas)with known answers(laws in our case). Measure retrieval accuracy, answer relevance, and whether the correct source documents are mentioned with right citations.

3. Ai based thermal design recommendation [what I learned from exploring TASK 1]

Problem: choosing the right heat sink dimensions, material, airflow configuration does require multiple simulation and iterations and good engineering experience. Ai can suggest design improvements based on patterns learned from previous successful designs which would help engineers reach an optimized solution faster.

Data required: past thermal simulation results, certain design parameters, material properties, cooling performance metrics , previous good working designs and many more.

AI type: Combination of Machine Learning (for pattern recognition) and genai (to explain why the recommendation is being made in easy language).

Validation: Compare AI recommendations with those made by experienced thermal engineers and validate them through simulation. We could also measure the improvements in temperature reduction and the decrease in the number of design iterations needed.

4. Financial Anomaly Detection [Related to my MLops project]

Project Post:

https://www.linkedin.com/posts/shlok-talhar-63ab80282_fintech-cicd-mlops-ugcPost-7448959894814150656-xmjY/

Problem: Financial institutions process millions of transactions every day, making it difficult for analysts to manually identify fraudulent activities or unusual spending patterns(like a person who never withdraws more than 10,000 at once , withdrawing 1,00,000 at a time). Ai can automatically detect suspicious transactions, account behavior, or payment patterns that deviate from normal activity, allowing them to be reviewed before financial loss occurs.

Data source: historical or past transaction records, customer spending patterns, merchant information, account activity, and previously confirmed fraud cases.

AI type: Machine Learning (anomaly detection) .

Validation: Compare detected anomalies against historical fraud cases and measure precision, recall,etc .. also evaluate how many fraudulent transactions are detected while minimizing unnecessary alerts for legit customers.

